Sanjay Kumar Bairi

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SUMMARY

Highly motivated, disciplined, Results-driven Data analyst, Data Engineer, business analyst with 3+ years of hands-on experience designing and optimizing scalable data solutions. Had the ability to learn new programming languages or techniques quickly. Experienced in architecting data pipelines, optimizing SQL workloads, and developing predictive analytics solutions that transform complex datasets into actionable business insights. Proven success in improving data quality, automating manual processes, and delivering measurable business value through cloud-based analytics and reporting solutions.

Hands-on expertise with AWS services including S3, Glue, Lambda, Athena, Redshift, and SageMaker, along with Databricks, Apache Spark, PySpark, Airflow, and dbt for large-scale data processing and orchestration.

Strong background collaborating with cross-functional stakeholders to gather requirements, design scalable data architectures, implement data governance best practices, and translate business needs into high-quality analytical solutions. Passionate about leveraging modern cloud technologies, automation, and machine learning to solve complex business challenges.

SKILLS AND TOOLS

Languages

- Java, Python(Keras,Tensorflow,PyTorch,numpy,pandas,matplotlib)

Skills

- Exploratory Data Analysis (EDA), Detailed Trend Analysis,
- Developing ETL/ELT pipelines.
- Cloud Platforms: AWS (Glue, Lambda, DynamoDB, Redshift, S3),
- Azure SQL Database, Azure Data Factory
- Data Warehousing: Snowflake, Redshift
- Data Orchestration: Apache Airflow, dbt
- Big Data & Analytics: Databricks, Apache Spark, PySpark,
- SQL, MS-SQL, Oracle SQL, PL/SQL, PostgreSQL,
- Data Visualization: Tableau, Power BI, Pandas, Seaborn, Matplotlib
- Containerization & Streaming: Docker, Kafka
- Application UI Design: Angular, Figma, HTML, CSS, JavaScript,
- TypeScript, Bootstrap, Nodejs, ExpressJS,PHP,.net.

Maintain data integrity.
Complex Data Interpretation.
Data Profiling, Data Modeling
Data Ingestions, Salesforce CRM.
Data Governance and Compliance.

EXPERIENCE

Data Analyst | Smart internz **Jan 2022-Jan 2023**

- Designed analytical data models using Star and Snowflake Schemas, improving reporting query performance.
- Developed and automated cloud-based ETL/ELT pipelines using Python, SQL, AWS Glue, and Amazon S3, integrating data from multiple relational and flat-file sources into centralized analytical datasets, reducing manual data preparation effort by 70%.
- Architected and delivered 15+ enterprise financial dashboards using Power BI, Tableau, and Amazon QuickSight, leveraging DAX, Power Query, drill-through reports, row-level security (RLS), KPI scorecards, and dynamic filtering, improving executive decision-making efficiency by 40%.
- Implemented comprehensive data quality frameworks using Python, SQL, and AWS Glue Data Quality, performing automated validation, reconciliation, duplicate detection, null analysis, and business rule enforcement to maintain 99% data accuracy and ensure reliable financial reporting.
- Partnered with finance, operations, and executive stakeholders to gather business requirements, perform data mapping and gap analysis, and translate complex business problems into scalable analytical solutions using SQL, Python, and AWS analytics services, enabling data-driven strategic planning.
- Optimized large-scale SQL workloads across Amazon RDS/PostgreSQL by implementing indexing strategies, query refactoring, execution plan analysis, partitioning, and materialized views, reducing average query execution time by 45% and improving dashboard refresh performance from 8 minutes to under 2 minutes for 50+ concurrent users.
- Built cloud-native data pipelines using AWS Lambda, Amazon S3, AWS Glue, and Amazon EventBridge to automate ingestion, transformation, and loading of financial and operational datasets into reporting environments, eliminating manual intervention and improving pipeline reliability.
- Performed advanced trend analysis, forecasting, and statistical modeling using Python (Pandas, NumPy, SciPy, Matplotlib) on datasets stored in Amazon S3 and queried through Amazon Athena, identifying seasonal demand patterns and forecasting inventory requirements that reduced excess inventory costs price through optimized stock planning.

Data Scientist | IBM **Jan 2022-Jan 2023**

- Engineered scalable end-to-end machine learning pipelines using Python (Pandas, NumPy, Scikit-learn), SQL, and AWS SageMaker, automating data ingestion, feature engineering, preprocessing, feature selection, and model training for datasets containing millions of records, reducing data preparation time by 75% while improving feature quality by 25%
- Developed and automated cloud-based ETL/ELT pipelines using Python, SQL, AWS Glue, and Amazon S3, integrating data from multiple relational and flat-file sources into centralized analytical datasets, reducing manual data preparation effort by 70%.
- Architected and delivered 15+ enterprise financial dashboards using Power BI, Tableau, and Amazon QuickSight, leveraging DAX, Po
- Designed, trained, and optimized 8+ supervised machine learning models including Logistic Regression, Random Forest, XGBoost, Support Vector Machines (SVM), and Gradient Boosting, utilizing hyperparameter tuning (GridSearchCV, RandomizedSearchCV) to achieve 85–92% classification accuracy across customer churn prediction, fraud detection, demand forecasting, and risk assessment use cases.
- Developed interactive machine learning analytics dashboards using Power BI, Tableau, and Amazon QuickSight, integrating prediction outputs, feature importance (SHAP), confusion matrices, ROC-AUC curves, precision-recall metrics, and real-time KPI monitoring to communicate model performance and business impact to executive stakeholders.
- Conducted comprehensive Exploratory Data Analysis (EDA) and statistical analysis using Pandas, SciPy, StatsModels, and Matplotlib, performing correlation analysis, feature importance evaluation, hypothesis testing, multivariate regression, outlier detection, and A/B testing to identify key business drivers and validate model assumptions.
- Built and deployed production-ready machine learning inference pipelines using AWS SageMaker, AWS Lambda, Amazon S3, API Gateway, and Docker, exposing RESTful prediction APIs and implementing automated retraining workflows triggered by new data arrivals, maintaining 87%+ production accuracy over a 12-month lifecycle.
- Implemented robust model validation frameworks utilizing Stratified K-Fold Cross Validation, Holdout Testing, Precision-Recall analysis, ROC-AUC evaluation, F1-score optimization, and confusion matrix analysis, ensuring model generalization, minimizing overfitting, and producing reproducible results documented for data science and business leadership.
- Optimized machine learning data pipelines through advanced SQL query optimization, feature engineering, missing value imputation, categorical encoding, normalization, and dimensionality reduction techniques, improving training efficiency by 40% and reducing model execution time for large-scale datasets.

Student Analyst | Intern- Vaagdevi Engineering College **Aug 2020- May 2023**

- Developed and maintained full-stack web applications for academic and departmental projects using HTML5, CSS3, JavaScript, Python, and Node.js, implementing responsive user interfaces and backend functionality.
- Designed and implemented interactive front-end components with HTML, CSS, Bootstrap, and JavaScript, improving application usability and ensuring compatibility across multiple browsers and devices.
- Built backend modules and RESTful API endpoints using Python/Flask and Node.js, enabling seamless communication between the application, database, and user interface.
- Developed and optimized SQL queries for CRUD operations, data retrieval, validation, and reporting using MySQL/PostgreSQL, ensuring data integrity and efficient database performance.
- Assisted in database design by creating tables, defining relationships, and implementing normalization techniques to improve data organization and maintainability.
- Performed application testing, debugging, and bug fixes using browser developer tools and IDE debugging features, reducing defects and improving overall application stability.
- Collaborated with faculty members and student teams to gather project requirements, translate functional specifications into technical solutions, and deliver project milestones within deadlines.Collaborated with faculty teams to troubleshoot issues, optimize performance, and deliver working prototypes on time Student Technical Intern – Vaagdevi Engineering College.

College Club

- Built a fully functional Campus Event Management App in **24 hours**, enabling student registration, event scheduling, and real-time event updates using HTML, CSS, JavaScript, Python/Node.js, and a lightweight database.
- Designed and developed a responsive, mobile-friendly user interface that improved usability and allowed seamless event registration across devices.
- Developed CRUD functionality for creating, updating, deleting, and managing events, registrations, and participant records.
- Designed the backend architecture and RESTful APIs to support efficient communication between the frontend and database.
- Optimized database queries to ensure fast retrieval of event schedules and registration details with minimal latency.
- Integrated automated registration confirmation and attendee tracking features to improve event management efficiency.
- Applied Git version control and collaborated effectively under tight hackathon deadlines using an agile development approach.
- Performed end-to-end testing, debugging, and issue resolution to ensure application stability before project submission.
- Demonstrated rapid problem-solving and full-stack development skills by delivering a production-ready prototype within a 24-hour development cycle.

EDUCATION

Master's in Information Systems | Auburn University at Montgomery, USA | GPA: 3.7 | **Jan 2024- Dec 2025**
Bachelors in Computer Science | Vaagdevi Engineering College, India |GPA 3.3 | **Aug 2019- Jul 2023**

Projects

E-Commerce Business Intelligence & Customer Analytics **GitHub | Portfolio**

- Analyzed 391K+ transactions, 4.2K customers, 3.6K products, generating $6.82M revenue.
- Conducted RFM segmentation, identifying Champions (940), Loyal Customers (983), At-Risk (766), and calculated $1.51M total CLV.
- Built Random Forest churn model (89% accuracy) and performed market basket analysis to guide cross-selling strategies.
- Developed interactive dashboards and revenue forecasts predicting $1.16M in 30-day revenue (+74% growth) to support strategic decisions.

Customer Churn Prediction System | Machine Learning **GitHub | Portfolio**
(Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn)

Built end-to-end ML classification system analyzing 7K+ telecom customer records to predict churn risk. Performed EDA, feature engineering, and model comparison (Logistic Regression, Random Forest, XGBoost). Created risk segmentation framework and delivered business recommendations with ROI analysis.

- Engineered ML churn prediction system using Logistic Regression, Random Forest, and XGBoost achieving 84% ROC-AUC; identified 87 high-risk customers and projected $21.8K annual savings with 251% ROI through data-driven retention strategies.

Retail Sales Analysis System | SQL & Business Intelligence **GitHub | Portfolio**
(SQL (SQLite), Python, Pandas, Matplotlib, Seaborn)

Built SQL-focused analytics platform analyzing 9,994 retail orders using advanced querying (window functions, CTEs, complex joins). Performed RFM customer segmentation, profitability analysis, and seasonal trend identification. Delivered actionable recommendations with quantified financial impact.

- Developed SQL analytics system with 60+ advanced queries (window functions, CTEs, joins) analyzing 9,994 orders; performed RFM segmentation identifying top 10% customers generating 50%+ of $2.3M revenue and delivered targeted retention strategies worth $150K annually

Real-Time Weather Data Pipeline & Notification System **Portfolio**
(Python, Apache Airflow, AWS (S3, Lambda, SES, IAM), OpenWeather API)

- Deployed automated ETL pipeline extracting weather forecasts from OpenWeather API, reducing data collection.
- Built event-driven architecture using AWS Lambda triggers on S3 bucket writes, enabling instant user notifications when weather thresholds exceeded
- Orchestrated scheduled data ingestion using Apache Airflow DAGs, ensuring 99.5% uptime and consistent data availability for downstream analysis
- Applied AWS security best practices with IAM roles for least-privilege access, preparing data infrastructure for enterprise-scale analytics

CERTIFICATIONS

Google Advanced Data Analytics Certificate
Microsoft AI & ML Engineering Certificate
AWS Certified Cloud Practitioner Certificate
AWS Certified Data Engineer Certificate
Databricks Cloud Integration Certification Certificate
Databricks Lakehouse Fundamentals Certificate